

MSDM 5056 Network Modeling Labs and Project (2025)

There are two labs and one project in this course. Each of the labs is worth 17.5%, and the project is worth 25% of the final grade. For the labs, you only need to submit some figures with captions that explain how you obtain them and what they show. For the project, you need to write a short report and **discuss** the properties of the networks you construct. The deadline of the two labs is November 26, 2025, and the deadline of the project is December 10, 2025.

Lab 1: Random Network (*Refer to Lecture 6*)

Generate five random networks which respectively have $N = 100, 300, 900, 2700,$ and 8100 nodes. They all have an average degree $\langle k \rangle = 20$.

- Plot their degree distribution $P(k)$.
- Plot their average shortest distance L against $\ln N$. Comment on the relation between L and N . (Design your program wisely, otherwise it surely gets stuck at a large N .)

Lab 2: Small-World Network (*Refer to Lecture 7*)

Generate some small-world networks with 1000 nodes and the same average degree. Illustrate their crossover behavior by plotting their average shortest distance L and average clustering coefficient C against their rewiring probability p like Fig. 1.

You should average your measurement over a few samples for each p in order to get smooth curves. You must also report your sample size.

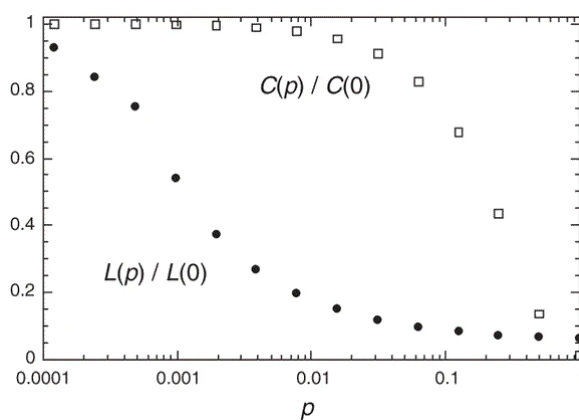


Fig. 1. Figure 2 from Watts and Strogatz [1].

Project

Choose two datasets from a reliable source. One should be small and contain only 100 to 500 nodes. The other one should be large and contain at least 1000 nodes. You must not use the datasets that the tutorials have used.

(a) Plot their distributions of degree, shortest distance, and clustering coefficient, then compute their respective average.

(b) Community Detection (*Refer to Lecture 5*)

- (i) Find the communities of the small network with the Girvan-Newman algorithm, then visualize the results.
- (ii) Find the communities of the large network with any algorithm you like, then visualize the results with a diagram in which
 - a node represents a community;
 - the size of a node indicates the size of a community;
 - the width of a node's border indicates the number of edges inside a community; and
 - the width of an edge between two nodes indicates the number of edges between two communities.

(c) Network Resilience (*Refer to Lecture 9*)

Using only the large network, perform (i) a random attack and (ii) a targeted attack by removing nodes from it. Plot the size of its giant component against the fraction of removed nodes like p.16–17 in Lecture 9.

For the random attack, you need to repeat your experiment at least a few times and take the average in order to get a smooth curve. For the targeted attack, you need to specify your strategy.

Write a short report to show all the answers you have obtained. Briefly discuss what kind of networks you have (by, e.g., analyzing their statistical and topological properties) and what other information you may get from your results.

You must write your report in an academic style. If you have no idea about this, you may use the format of Lecture Notes in Computer Science (LNCS), whose template you can download online.

References

1. Watts, D. J., Strogatz, S. H.: Collective dynamics of 'small-world' networks. *Nature* 393(6684), 440–442 (1998).